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Geometry and Cohomology in Group Theory

Edited by
Peter H. Kropholler, Graham A. Niblo
& Ralph Stöhr



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Edited by

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Preface

Refereed Conference Proceedings
The London Mathematical Society Symposium† on
Geometry and Cohomology in Group Theory
12–22 July 1994

The cross-fertilization of ideas from abstract group theory with those from geometry, the use of topological and cohomological techniques have contributed over recent years to revitalising group theory and representation theory in many exciting ways. Not only has this brought impressive mathematical advances, but again it has drawn in different kinds of specialists. At this London Mathematical Society Symposium we aimed to draw together experts in algebra, geometry, representation theory and cohomology to make a contribution to this interchange of ideas. It is an area where there is marked strength in the United Kingdom, traditionally the home of many group theorists, but nowadays also active in cohomological and geometric group theory.

The closing report to the Science and Engineering Research Council* began thus:

“The success of this meeting owed much to the strong list of participants, the high standard of lectures by invited speakers and to the good environment at Grey College which encouraged many collaborative research projects. The success can be measured by the strength of the articles which we are assembling for the Conference Proceedings.”

We thank all those authors whose articles were submitted for this volume, and also the equal number of referees. We emphasise that every effort was made

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* Now replaced by the Engineering and Physical Sciences and the Particle Physics and Astronomy Research Councils.

to have papers refereed to the same standard as academic and learned mathematical journals. There are a number of survey articles here including those by Carlson, Cornick, Grigorchuk, Linnell, Mikhajlovskii and Ol'shanskii, and Wilson. These cover ground from cohomology and representation theory, analytic methods and the use of von Neumann algebras, the application of hyperbolic groups and the structure of soluble groups. Davis gave a series of lectures on buildings. Further contributions were made following collaborations at the meeting, for example the paper by Neumann and Rowley.

Every participant contributed to the success of this meeting. For the smooth organization we are indebted to the hard work of Sue Nesbitt, Rachel Duke and Ruth Silverstone. The support of the London Mathematical Society and Tony Scholl were invaluable, and the accommodation at Grey College provided a comfortable and congenial atmosphere. The first named editor would like to thank Richard Platten for his assistance with authors' proofs.

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LMS Durham Symposium on: Geometry and Cohomology in Group Theory

Tuesday 12th July to Friday 22nd July 1994



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On the Cohomology of $SL_2(\mathbb{Z}[1/p])$

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0. Introduction

In this note we compute the integral cohomology of the discrete groups $SL_2(\mathbb{Z}[1/p])$, p a prime. According to Serre [6] these are groups of virtual cohomological dimension 2. The method we use is to exploit the fact that these groups can be expressed as an amalgamation of two copies of $SL_2(\mathbb{Z})$ along the subgroup $\Gamma_0(p)$ of 2×2 matrices with lower left hand entry divisible by p . We first compute the cohomology of this virtually free group (using a tree on which it acts with finite isotropy and compact quotient), and then use the well-known Mayer–Vietoris sequence in cohomology to obtain our result. We assume that p is an odd prime larger than 3. The cases $p = 2, 3$ must be treated separately; we discuss them at the end of the paper. We are grateful to the referee for his extremely useful remarks, and to J.-P. Serre for pointing out Proposition 3.1.

1. Double cosets and permutation modules

In this section we calculate certain double coset decompositions which will play a key rôle in our approach. Let $G = SL_2(\mathbb{F}_p)$, and $B \subset G$ the subgroup consisting of all matrices with lower left hand entry equal to zero. It is easy to see that $B \cong \mathbb{Z}/p \times_T \mathbb{Z}/p - 1$, a semidirect product. We will denote by C_2 , C_4 and C_6 the cyclic subgroups generated by the following three respective matrices of orders 2, 4 and 6:

$$a_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad a_4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad a_6 = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

It is easy to check that the set of right cosets $B \backslash G$ decomposes as follows:

$$B \backslash G = Ba_4 \sqcup \left(\bigsqcup_{x \in \mathbb{F}_p} B \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} \right).$$

Now as a_2 is central, we obtain a double coset decomposition for G using B and C_2 as follows:

$$G = Ba_4C_2 \sqcup \left(\bigsqcup_{x \in \mathbb{F}_p} B \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} C_2 \right).$$

Applying the usual induction restriction formula we obtain

$$\mathbb{Z}[G/C_2] \big|_B \cong (\mathbb{Z}[B/C_2])^{p+1}. \quad (1.1)$$

Next we consider the double cosets using C_4 . Note that we have

$$\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -x \end{pmatrix}$$

and

$$\begin{pmatrix} 0 & -1 \\ 1 & -x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1/x & 1 \end{pmatrix} = \begin{pmatrix} -1/x & -1 \\ 0 & -x \end{pmatrix}.$$

From this we conclude that if $x \neq -1/x$ or 0 , then

$$B \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = B \begin{pmatrix} 1 & 0 \\ -1/x & 1 \end{pmatrix}$$

and so

$$B \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} C_4 = B \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} \sqcup B \begin{pmatrix} 1 & 0 \\ -1/x & 1 \end{pmatrix}, \text{ and } Ba_4C_4 = B \sqcup Ba_4.$$

In each case the two cosets are permuted by the matrix a_4 . In the case when $x^2 + 1 = 0$, then the coset is fixed under this action, and hence the associated coset is equal to the double coset. It is an elementary fact that the polynomial $t^2 + 1$ will have roots in $\mathbb{F}_p$ (necessarily two distinct ones) if and only if $p \equiv 1 \pmod{4}$. Using this and the induction restriction formula yields

$$\mathbb{Z}[G/C_4] \big|_B \cong \mathbb{Z}[B/s_1C_4s_1^{-1}] \oplus \mathbb{Z}[B/s_2C_4s_2^{-1}] \oplus (\mathbb{Z}[B/C_2])^{(p-1)/2}$$

if $p \equiv 1 \pmod{4}$, and

$$\mathbb{Z}[G/C_4] \big|_B \cong (\mathbb{Z}[B/C_2])^{(p+1)/2} \quad \text{otherwise.} \quad (1.2)$$

The elements s_1, s_2 correspond to the two roots of the polynomial.

For C_6 , we must look at the action of the matrix of order 3, $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$ on the double cosets. In this case the orbit of the action will be of the form

$$B \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}, B \begin{pmatrix} 1 & 0 \\ 1/(1-x) & 1 \end{pmatrix}, B \begin{pmatrix} 1 & 0 \\ (x-1)/x & 1 \end{pmatrix}$$

provided $x \neq 1$. A coset will be fixed if and only if $x^2 - x + 1 = 0$; given that $p > 3$, this will have roots in $\mathbb{F}_p$ (necessarily two distinct ones) if and only if $p \equiv 1 \pmod{3}$. As for $x = 1$, the corresponding coset gives rise to the singular orbit

$$B, B \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, Ba_4.$$

From this we can deduce the following decomposition (notation as before):

$$\mathbb{Z}[G/C_6] |_B \cong \mathbb{Z}[B/s_1 C_6 s_1^{-1}] \oplus \mathbb{Z}[B/s_2 C_6 s_2^{-1}] \oplus (\mathbb{Z}[B/C_2])^{(p-1)/3}$$

if $p \equiv 1 \pmod{3}$ and

$$\mathbb{Z}[G/C_6] |_B \cong (\mathbb{Z}[B/C_2])^{(p+1)/3} \quad \text{otherwise.} \quad (1.3)$$

2. The cohomology of $\Gamma_0(p)$

The subgroup $\Gamma_0(p) \subset SL_2(\mathbb{Z})$ is defined by

$$\Gamma_0(p) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \mid c \equiv 0 \pmod{p} \right\}.$$

If $\Gamma(p)$ denotes the level p congruence subgroup, then clearly $\Gamma_0(p)$ can be expressed as an extension

$$1 \rightarrow \Gamma(p) \rightarrow \Gamma_0(p) \rightarrow B \rightarrow 1.$$

Recall [6] that $SL_2(\mathbb{Z})$ acts on a tree T with finite isotropy, and quotient a single edge. As $\Gamma(p)$ is torsion free, it acts freely on this tree, and so $G = SL_2(\mathbb{F}_p)$ acts on the finite graph $T/\Gamma(p)$. The isotropy subgroups of this action are precisely C_4 and C_6 for the vertices and C_2 for the edge. Let EB denote the universal B -space, then clearly using the projection $\pi : \Gamma_0(p) \rightarrow B$, $\Gamma_0(p)$ can be made to act diagonally on $EB \times T$, with trivial isotropy. As this space is contractible, its quotient under this action has the same homotopy type as the classifying space $B\Gamma_0(p)$, and so we have $B\Gamma_0(p) \cong EB \times_B T/\Gamma(p)$.

Let C^* denote the cellular cochains on the B -CW complex $T/\Gamma(p)$; then it is clear from the above that

$$C^0 \cong \mathbb{Z}[G/C_4] \downarrow_B \oplus \mathbb{Z}[G/C_6] \downarrow_B \quad \text{and that} \quad C^1 \cong \mathbb{Z}[G/C_2] \downarrow_B.$$

In this situation, there is a spectral sequence converging to $H^*(\Gamma_0(p), \mathbb{Z})$ (see [2]) with $E_1^{p,q} \cong H^q(B, C^p)$, which degenerates into a long exact sequence:

$$\dots \rightarrow H^i(\Gamma_0(p), \mathbb{Z}) \rightarrow H^i(B, C^0) \rightarrow H^i(B, C^1) \rightarrow H^{i+1}(\Gamma_0(p), \mathbb{Z}) \rightarrow \dots \quad (2.1)$$

where the middle arrow is induced by the coboundary map δ on C^* .

As a first application of the long exact sequence (2.1) we obtain

Proposition 2.2. *Under the above conditions, $H^1(\Gamma_0(p), \mathbb{Z}) \cong (\mathbb{Z})^{N(p)}$ where*

$$N(p) = \begin{cases} (p-7)/6, & \text{if } p \equiv 1 \pmod{12}; \\ (p+1)/6, & \text{if } p \equiv 5 \pmod{12}; \\ (p-1)/6, & \text{if } p \equiv 7 \pmod{12}; \\ (p+7)/6, & \text{if } p \equiv 11 \pmod{12}. \end{cases}$$

Proof The sequence (2.1) starts as

$$0 \rightarrow \mathbb{Z} \rightarrow (C^0)^B \rightarrow (C^1)^B \rightarrow H^1(\Gamma_0(p), \mathbb{Z}) \rightarrow H^1(B, C^0).$$

Recall that $H^1(B, C^0) = 0$, as C^0 is a permutation module. Hence calculating ranks completes the proof.

To compute the remaining cohomology groups we first switch to the associated projective group $P\Gamma_0(p)$. Note that there will be a situation analogous to that for the original group, except that throughout we must divide out by the central $\mathbb{Z}/2$. Note that if PB is the associated group for B , then C^1 will now be a free PB -module. Hence the corresponding long exact sequence degenerates to yield the isomorphism $H^{2i}(P\Gamma_0(p), \mathbb{Z}) \cong H^2(PB, C^0)$ for all $i \geq 0$, and the fact that all its odd dimensional cohomology (except in dimension 1) is zero. This is summarized in

Proposition 2.3. *For any integer $i \geq 1$, we have that*

$$H^{2i}(P\Gamma_0(p), \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/6 \oplus \mathbb{Z}/6, & \text{if } p \equiv 1 \pmod{12}; \\ \mathbb{Z}/2 \oplus \mathbb{Z}/2, & \text{if } p \equiv 5 \pmod{12}; \\ \mathbb{Z}/3 \oplus \mathbb{Z}/3, & \text{if } p \equiv 7 \pmod{12}; \\ 0, & \text{if } p \equiv 11 \pmod{12}. \end{cases}$$

and $H^{2i+1}(P\Gamma_0(p), \mathbb{Z}) = 0$.

□

Note that $H^1(P\Gamma_0(p), \mathbb{Z}) \cong H^1(\Gamma_0(p), \mathbb{Z})$.

Next we apply the spectral sequence over $\mathbb{Z}$ associated to the central extension

$$1 \rightarrow C_2 \rightarrow \Gamma_0(p) \rightarrow P\Gamma_0(p) \rightarrow 1.$$

Note that the interesting cases are if $p \equiv 1, 5 \pmod{12}$, and that the 3-torsion plays no role. As the group has periodic cohomology and is virtually free, it suffices to compute H^2 and H^3 . In total degree 2 we simply have the contributions from $H^2(C_2, \mathbb{Z}) \cong \mathbb{Z}/2$ and $H^2(P\Gamma_0(p), \mathbb{Z})_{(2)} \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2$. As 4-torsion must appear (there will be a subgroup of that order), we conclude that $H^2(\Gamma_0(p), \mathbb{Z})_{(2)} \cong \mathbb{Z}/4 \oplus \mathbb{Z}/2$.

In total degree 3, we only have one term

$$H^1(P\Gamma_0(p), H^2(C_2, \mathbb{Z})) \cong (\mathbb{Z}/2)^{N(p)+2}.$$

However, it is not hard to see that the map induced by the quotient in cohomology, $H^4(P\Gamma_0(p), \mathbb{Z}) \rightarrow H^4(\Gamma_0(p), \mathbb{Z})$ must be zero. This can be proved by comparing the two long exact sequences described above and using the corresponding fact for the map induced by the quotient $\mathbb{Z}/4 \rightarrow \mathbb{Z}/2$. The only possible differential on this horizontal edge group is $d_3 : E_3^{1,2} \rightarrow E_3^{4,0}$, hence it must have an image of 2-primary rank 2. We conclude that $E_\infty^{1,2} \cong (\mathbb{Z}/2)^{N(p)} \cong H^3(\Gamma_0(p), \mathbb{Z})$, and we obtain

Theorem 2.4. *For any integer $i \geq 1$, we have*

$$H^{2i}(\Gamma_0(p), \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/12 \oplus \mathbb{Z}/6, & \text{if } p \equiv 1 \pmod{12}; \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & \text{if } p \equiv 5 \pmod{12}; \\ \mathbb{Z}/3 \oplus \mathbb{Z}/6, & \text{if } p \equiv 7 \pmod{12}; \\ \mathbb{Z}/2, & \text{if } p \equiv 11 \pmod{12} \end{cases}$$

and

$$H^{2i+1}(\Gamma_0(p), \mathbb{Z}) \cong (\mathbb{Z}/2)^{N(p)}.$$

□

3. Calculation of the cohomology

To begin we recall that aside from the natural inclusion we also have an injection $\rho : \Gamma_0(p) \rightarrow SL_2(\mathbb{Z})$, given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & pb \\ p^{-1}c & d \end{pmatrix}.$$

Using these two imbeddings, we can construct the amalgamated product [6]

$$SL_2(\mathbb{Z}[1/p]) \cong SL_2(\mathbb{Z}) *_{\Gamma_0(p)} SL_2(\mathbb{Z}).$$

In addition we have that

$$\overline{H}^r(SL_2(\mathbb{Z}), \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/12 & \text{if } r \text{ is even;} \\ 0 & \text{if } r \text{ is odd.} \end{cases}$$

We can identify ρ^* with the ordinary restriction map.

Using the Mayer–Vietoris sequence associated to an amalgamated product we see that $H^1(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) = 0$ and that we have exact sequences

$$0 \rightarrow \mathbb{Z}^{N(p)} \rightarrow H^2(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \rightarrow \mathbb{Z}/12 \oplus \mathbb{Z}/12 \rightarrow H^2(\Gamma_0(p), \mathbb{Z}) \rightarrow H^3(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \rightarrow 0$$

and

$$0 \rightarrow (\mathbb{Z}/2)^{N(p)} \rightarrow H^{2i}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \rightarrow \mathbb{Z}/12 \oplus \mathbb{Z}/12 \rightarrow H^{2i}(\Gamma_0(p), \mathbb{Z}) \rightarrow H^{2i+1}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \rightarrow 0.$$

The cohomology will evidently be 2-fold periodic above dimension 2, which is in fact the virtual cohomological dimension of $SL_2(\mathbb{Z}[1/p])$.

We will need the following result, which is due to J.-P. Serre [7]. It can also be proved using an explicit presentation for the group, described in [1].

Proposition 3.1.

$$H_1(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/3 & \text{if } p=2; \\ \mathbb{Z}/4 & \text{if } p=3; \\ \mathbb{Z}/12 & \text{otherwise.} \end{cases}$$

□

Hence we have that for $p > 3$, $H^2(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong (\mathbb{Z})^{N(p)} \oplus \mathbb{Z}/12$.

Let $A(p)$ denote the number $12/|Q(p)|$, where $Q(p)$ is the largest cyclic subgroup in $H^2(\Gamma_0(p), \mathbb{Z})$. Then, from the fact that the restriction from the cohomology of $SL_2(\mathbb{Z})$ to that of its cyclic subgroups factors through $\Gamma_0(p)$, we deduce that the sequence above simplifies to yield

$$0 \rightarrow (\mathbb{Z}/2)^{N(p)} \rightarrow H^{2i}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \rightarrow \mathbb{Z}/12 \oplus \mathbb{Z}/A(p) \rightarrow 0 \quad (3.2)$$

and

$$H^{2i+1}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong H^{2i}(\Gamma_0(p), \mathbb{Z})/Q(p).$$

Moreover, from our previous calculation for $\Gamma_0(p)$, we have that

$$Q(p) = \begin{cases} \mathbb{Z}/12 & \text{if } p \equiv 1 \pmod{12}; \\ \mathbb{Z}/4 & \text{if } p \equiv 5 \pmod{12}; \\ \mathbb{Z}/6 & \text{if } p \equiv 7 \pmod{12}; \\ \mathbb{Z}/2 & \text{if } p \equiv 11 \pmod{12}. \end{cases}$$

It remains only to determine precisely what this extension (3.2) looks like. We need only be concerned with the 2-primary component. Recall that $H^{2i}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z})$ is a quotient of $H^2(SL_2(\mathbb{Z}[1/p]), \mathbb{Z})$, as $SL_2(\mathbb{Z}[1/p])$ is a group of virtual cohomological dimension 2, which has 2-fold periodic cohomology. This can also be explained by saying that the sequence in high even dimensions can be identified with the corresponding sequence in 2-dimensional Farrell cohomology (see [2]). Mapping one sequence into the other, we see that a $\mathbb{Z}/4$ summand must split off for all values of $p > 3$. This means that the sequence *will split* for $p \equiv 1, 5 \pmod{12}$. For the remaining cases $p \equiv 7, 11 \pmod{12}$, it remains to solve the extension problem after splitting off the $\mathbb{Z}/4$ summand. However, from our knowledge of H^2 , we know that the dimension of $H^{2i} \otimes \mathbb{Z}/2$ can be at most $N(p) + 1$. We infer that the reduced extension *does not split*, and must necessarily have a $\mathbb{Z}/4$ summand present. Hence we have the following complete calculation:

Theorem 3.3. *Let p be an odd prime larger than 3, then*

$$H^1(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong 0$$

and

$$H^2(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{(p-7)/6} \oplus \mathbb{Z}/12 & \text{if } p \equiv 1 \pmod{12}; \\ \mathbb{Z}^{(p+1)/6} \oplus \mathbb{Z}/12 & \text{if } p \equiv 5 \pmod{12}; \\ \mathbb{Z}^{(p-1)/6} \oplus \mathbb{Z}/12 & \text{if } p \equiv 7 \pmod{12}; \\ \mathbb{Z}^{(p+7)/6} \oplus \mathbb{Z}/12 & \text{if } p \equiv 11 \pmod{12}. \end{cases}$$

For $i \geq 2$, we have

$$H^{2i}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong \begin{cases} (\mathbb{Z}/2)^{(p-7)/6} \oplus \mathbb{Z}/12 & \text{if } p \equiv 1 \pmod{12}; \\ (\mathbb{Z}/2)^{(p+1)/6} \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/3 & \text{if } p \equiv 5 \pmod{12}; \\ (\mathbb{Z}/2)^{(p-7)/6} \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/4 & \text{if } p \equiv 7 \pmod{12}; \\ (\mathbb{Z}/2)^{(p+1)/6} \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/12 & \text{if } p \equiv 11 \pmod{12}, \end{cases}$$

and

$$H^{2i-1}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/6 & \text{if } p \equiv 1 \pmod{12}; \\ \mathbb{Z}/2 & \text{if } p \equiv 5 \pmod{12}; \\ \mathbb{Z}/3 & \text{if } p \equiv 7 \pmod{12}; \\ 0 & \text{if } p \equiv 11 \pmod{12}. \end{cases}$$

□

Of the two remaining cases ($p = 2, 3$) the second one can be done in a manner totally analogous to what we have presented. Specifically we have that

$$H^i(\Gamma_0(3), \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } i = 0, 1; \\ \mathbb{Z}/6 & \text{if } i \text{ is even}; \\ \mathbb{Z}/2 & \text{if } i > 1 \text{ is odd}. \end{cases}$$

We obtain that $H^2(SL_2(\mathbb{Z}[1/3]), \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/4$, $H^{2i+1}(SL_2(\mathbb{Z}[1/3]), \mathbb{Z}) \cong 0$, and it only remains to deal with the extension

$$0 \rightarrow \mathbb{Z}/2 \rightarrow H^{2i}(SL_2(\mathbb{Z}[1/p]), \mathbb{Z}) \rightarrow \mathbb{Z}/12 \oplus \mathbb{Z}/2 \rightarrow 0.$$

As before the $\mathbb{Z}/12$ must split off, and by rank considerations an extra $\mathbb{Z}/4$ summand must appear. To summarize, we have

$$H^i(SL_2(\mathbb{Z}[1/3]), \mathbb{Z}) \cong \begin{cases} 0 & \text{if } i \text{ is odd}; \\ \mathbb{Z} \oplus \mathbb{Z}/4 & \text{if } i = 2; \\ \mathbb{Z}/12 \oplus \mathbb{Z}/4 & \text{if } i = 2j, j > 1. \end{cases}$$

The case $p = 2$ is complicated by the fact that $\Gamma(2)$ is not torsion-free. However, one can still make use of the associated projective groups; $P\Gamma(2)$ is free of rank 2, and there is an extension

$$1 \rightarrow P\Gamma(2) \rightarrow P\Gamma_0(2) \rightarrow \mathbb{Z}/2 \rightarrow 1.$$

Analyzing the action on the corresponding graph, it is not hard to show that

$$H^*(P\Gamma_0(2), \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } i \text{ is 0 or 1}; \\ \mathbb{Z}/2 & \text{if } i \text{ is even}; \\ 0 & \text{otherwise}. \end{cases}$$

Then, using the central extension, it is direct to show that

$$H^i(\Gamma_0(2), \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } i \text{ is 0 or 1}; \\ \mathbb{Z}/4 & \text{if } i \text{ is even}; \\ \mathbb{Z}/2 & \text{if } i \text{ is odd, } i > 1. \end{cases}$$

Using the Mayer–Vietoris sequence as before, we obtain that

$$H^2(SL_2(\mathbb{Z}[1/2]), \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/3, H^i(SL_2(\mathbb{Z}[1/2]), \mathbb{Z}) = 0$$

if i is odd, and a short exact sequence

$$0 \rightarrow \mathbb{Z}/2 \rightarrow H^{2i}(SL_2(\mathbb{Z}[1/2]), \mathbb{Z}) \rightarrow \mathbb{Z}/12 \oplus \mathbb{Z}/3 \rightarrow 0.$$

In this case, we do not know that the $\mathbb{Z}/12$ summand must split off. Looking at the 2-primary part, we see that it can have at most one cyclic summand. The only possibility is $\mathbb{Z}/8$, and we have

$$H^i(SL_2(\mathbb{Z}[1/2]), \mathbb{Z}) \cong \begin{cases} 0 & \text{if } i \text{ is odd;} \\ \mathbb{Z} \oplus \mathbb{Z}/3 & \text{if } i = 2; \\ \mathbb{Z}/24 \oplus \mathbb{Z}/3 & \text{if } i = 2j, j > 1. \end{cases}$$

Remarks. Note that this last group has no finite subgroups of order eight, which makes its cohomology rather interesting. We are grateful to Hans–Werner Henn for pointing out the correct cohomology of this group. Also we would like to point out that Naffah [4] has calculated the 3-adic component of the Farrell cohomology $\widehat{H}^*(SL_2(\mathbb{Z}[1/N]), \mathbb{Z})$ for any integer N . This of course can be used to recover our calculations at $p = 3$ in dimensions larger than 2. Moss [5] has computed the rational cohomology of $SL_2(\mathbb{Z}[1/N])$ which again can be used to recover part of our results. The general calculation of $H^*(SL_2(\mathbb{Z}[1/N]), \mathbb{Z})$ seems to be a rather complicated but interesting open problem. We refer to [3] for more on this.

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Cohomology of Sporadic Groups, Finite Loop Spaces, and the Dickson Invariants

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1. Introduction

Whatever one may think of a proof that covers over ten thousand pages of journal articles, some of which have still not appeared in print, the classification of finite simple groups is a remarkable theorem. It says that (apart from the cyclic groups of prime order and the alternating groups) most finite simple groups are groups of Lie type. These are the analogues of the compact Lie groups, defined over a finite field. They admit a uniform description in terms of the fixed points of certain automorphisms related to the Frobenius map, on the corresponding algebraic groups in prime characteristic. The classifying space of a finite group of Lie type fibers over that of the corresponding Lie group with fibers which are cohomologically finite away from the defining characteristic. Apart from the groups of Lie type, there are the alternating groups A_n ($n \geq 5$), and twenty-six other groups called the sporadic simple groups.

The first five sporadic groups were discovered by Mathieu in the late nineteenth century. The remaining twenty-one were discovered in the nineteen sixties and seventies. The largest is the Fischer–Griess Monster, which has order roughly 8×10^{53} . For a wealth of information on the sporadic groups and other “small” finite simple groups, the reader is referred to the ATLAS of finite groups [Atlas]. A great deal of effort has gone into trying to understand these sporadic groups. For example, the “Monstrous Moonshine” [CN] (see also [CS]) is a remarkable series of observations, still not fully understood, connecting various of the sporadic groups, especially those involved in the Monster, with the theory of modular functions. This article is about another

attempt to understand some of the sporadic groups from a different point of view, namely that of the topology of their classifying spaces.

The story started with the construction by Bill Dwyer and Clarence Wilkerson [DW2] of a new finite loop space at the prime two, which they call $DI(4)$. The mod two cohomology of its classifying space is equal to the rank four Dickson invariants, whence the name. Finite loop spaces are sufficiently like compact Lie groups that this prompted the question of what finite simple groups could be associated to them in the way that the finite groups of Lie type are associated to the compact Lie groups.

The first group that was considered in this regard was Conway's group Co_3 , which has a 2-local structure closely associated with the 2-local structure of $DI(4)$. It turns out that the classifying space of Co_3 fibers over the classifying space of $DI(4)$ with fibers cohomologically finite at the prime two. However, this turns out to be more akin to the relation between M_{12} and G_2 than to the relation between $G_2(q)$ and G_2 .

The finite simple groups which should be regarded as the groups "of Lie type $DI(4)$ " over the field of q elements don't literally exist as groups. They were first considered by Ron Solomon [So] in an attempt to prove that Co_3 is determined by a Sylow 2-subgroup. He was led to consider simple groups in which the centralizer of an involution was $Spin(q)$ (q odd), possibly with some decoration of odd order. In order to prove nonexistence, Solomon was obliged to write down what appeared to be a consistent fusion pattern for the Sylow 2-subgroup, and then pass to an odd prime to obtain a contradiction. It turns out that despite the nonexistence of these groups, they have perfectly good 2-completed classifying spaces, which map to the classifying space of $DI(4)$ with fibers which are cohomologically finite at the prime two.

This prompts the speculation that there should exist a theory of " p -local groups", in which one only gives a Sylow p -subgroup and a fusion pattern. The fusion pattern should obey a set of axioms which are strong enough to be able to build a p -completed classifying space. One possible candidate for such a set of axioms has been written down by Puig [Pu] in connection with his attempts to formalise the local theory of blocks, but much work remains to be done in this area.

2. Classifying spaces

We begin by describing the theory of classifying spaces, and explain why they are rigid enough that new maps between them should be considered to be interesting.

For any topological group G , there exists a contractible space EG together with a free action of G on EG . A uniform construction was found by Milnor [Mi]. Roughly speaking, the idea is to form an infinite join $G * G * \cdots$ of

copies of G as a topological space, and let G act diagonally. The join $X * Y$ of two spaces X and Y consists of the space obtained from $X \times I \times Y$ by identifying $(x, 0, y)$ with $(x', 0, y)$ for all $x, x' \in X$ and $y \in Y$, and $(x, 1, y)$ with $(x, 1, y')$ for all $x \in X$ and $y, y' \in Y$. The difficult point in Milnor's construction is the exact description of the topology on the infinite join. However, we shall mostly be interested in the case where G is either discrete or a compact Lie group. In this case, we can just give the infinite join the weak topology with respect to the finite joins. Then EG is a CW-complex, so to see that it is contractible, it suffices to see that any map from the n -sphere S^n to EG extends to a map from the $(n+1)$ -disk D^{n+1} . Since S^n is compact, any such map has image contained in some finite join of copies of G , and then the next higher join contains the cone on this join, so that the map can be extended to D^{n+1} in this next join.

We write BG for the quotient of EG by the action of G . It turns out that the classifying space BG is determined up to homotopy by G , independently of the choice of contractible space EG . Thus if G is a subgroup of G' , then using EG' instead of EG , we see that there is an induced fibration $BG \rightarrow BG'$ with fiber the coset space G'/G . Similarly if N is a normal subgroup of G , we may use $EG \times E(G/N)$ instead of EG , with G acting on both factors, and obtain an induced fibration $BG \rightarrow B(G/N)$ with fiber BN . In this way, for any homomorphism of topological groups $\alpha : G \rightarrow G'$, we have an induced map of classifying spaces $B\alpha : BG \rightarrow BG'$, unique up to homotopy.

If G is discrete, BG is an Eilenberg–Mac Lane space $K(G, 1)$, and EG is its universal cover. In any case, the loop space ΩBG is homotopy equivalent to G as a topological space.

Classifying spaces are homotopically rather rigid. For example, if G and G' are discrete groups, then the set of homotopy classes of maps $[BG; BG']$ is in natural bijection with $\text{Hom}_{grp}(G, G')/\text{Inn}(G')$. Here, the group $\text{Inn}(G')$ of inner automorphisms of G' acts in the obvious way by composition on the set $\text{Hom}_{grp}(G, G')$ of group homomorphisms from G to G' . The bijection sends a map $\alpha : G \rightarrow G'$ to $B\alpha : BG \rightarrow BG'$. The inverse of this bijection sends a map $BG \rightarrow BG'$ to its effect on π_1 . This, of course, depends on a choice of basepoints, but the orbit under the action of $\text{Inn}(G')$ does not.

On the other hand, if G is a connected simple compact Lie group, then a theorem of Jackowski, McClure and Oliver (Theorem 3 in [JMO]) says that there is a natural bijection

$$\text{Hom}_{Liegrp}(G, G)/\text{Inn}(G) \wedge \{k \geq 0 : k = 0 \text{ or } (k, |W|) = 1\} \rightarrow [BG; BG]$$

sending (α, k) to $\Psi^k \circ B\alpha$. Here, the smash product denotes the quotient of the direct product given by identifying all elements in which at least one coordinate is zero. For $k > 0$, the operations Ψ^k are the unstable Adams psi

operations, which are known to exist for any connected simple compact Lie group and any integer k coprime to the order of the Weyl group W . They commute up to homotopy with maps coming from Lie group homomorphisms. If $k = 0$, Ψ^k is taken to be the identity map. The map Ψ^k is characterised by the property that its effect on degree $2i$ rational cohomology is multiplication by k^i for each $i \geq 0$.

These theorems indicate that it is of interest to construct maps of classifying spaces which do not come from group homomorphisms and Adams operations. Some examples for connected Lie groups can be found in [JMO]. For maps from the classifying space of a finite group to that of a compact Lie group, the situation is rather unclear, and it is possible that one should try to build a branch of representation theory based on such maps.

A theorem of Dwyer and Wilkerson [DW] indicates that we may work one prime at a time in this situation. Writing $X_p^\wedge$ for the Bousfield–Kan p -completion of a space X , their theorem states the following.

Theorem 2.1. *If π is a finite group and G is a connected compact Lie group then*

$$\mathrm{Map}(B\pi, BG) \simeq \prod_p \mathrm{Map}((B\pi)_p^\wedge, (BG)_p^\wedge).$$

Here, and from now on in this article, one must work with simplicial sets rather than topological spaces, and “Map” denotes the simplicial mapping space. The homotopy category of topological spaces is equivalent to the homotopy category of simplicial sets (these homotopy categories are formed by inverting weak equivalences), via singular simplices and geometric realization. So we use the term “space” from now on to mean “simplicial set”, and we replace a classifying space by its simplicial set of singular simplices.

The effect of Bousfield–Kan p -completion on the classifying space of a finite group G is as follows. Without changing the mod p cohomology, it gets rid of the mod l cohomology for every prime $l \neq p$. Its effect on the fundamental group G is to replace it by $G/O^p(G)$. So for example if G is a nonabelian finite simple group then the p -completion of BG is simply connected. The group G may not be recovered from the p -completion of BG , though the Sylow p -subgroup and the strong fusion of its subgroups may be recovered.

3. Dickson invariants

Let $\mathbb{F}_q$ be the finite field of q elements, with $q = p^a$ a prime power. Let $V = (\mathbb{F}_q)^n$ be an n -dimensional vector space over $\mathbb{F}_q$. Then the finite general linear group $G = GL_n(\mathbb{F}_q)$ acts on V in the obvious way, and hence on the coordinate ring

$$\mathbb{F}_q[V] = \mathbb{F}_q \oplus V^* \oplus S^2(V^*) \oplus \cdots$$

We regard this as a graded ring, with $\mathbb{F}_q$ in degree zero, $V^* = \text{Hom}_{\mathbb{F}_q}(V, \mathbb{F}_q)$ in degree one, $S^2(V^*)$ in degree two, and so on. It is a polynomial ring over $\mathbb{F}_q$ on generators forming an basis for V^* in degree one.

L. E. Dickson's theorem [Di] states that the invariants of this action also form a polynomial ring:

$$\mathbb{F}_q[V]^G = \mathbb{F}_q[c_{n,0}, \dots, c_{n,n-1}]$$

in generators $c_{n,i}$ (the Dickson invariants) of degree $q^n - q^i$. In fact, there is an explicit formula for $c_{n,i}$ as follows:

$$c_{n,i} = (-1)^{n-i} \sum_{\substack{W \subseteq V \\ \dim_{\mathbb{F}_q}(W)=i}} \prod_{\substack{\phi \in V^* \\ \phi|_W \neq 0}} \phi.$$

If you are a topologist, you may wish to double all the degrees in the above discussion, because if p is odd, elements of odd degree are supposed to anti-commute rather than commuting. Having done that, the polynomial ring on generators in degree two over $\mathbb{F}_p$ acquires an action of the Steenrod operations, since it is the mod p cohomology of a torus $T^n = \mathbb{R}^n/\mathbb{Z}^n = B\mathbb{Z}^n$ of rank n . The group $GL(n, \mathbb{Z})$ acts on T^n , and induces an action of the quotient group $GL(n, \mathbb{F}_p)$ on its mod p cohomology. This action commutes with the Steenrod operations, and so the invariants inherit an action of the Steenrod algebra.

Even if you are a topologist, in the case where $q = p = 2$, you may also wish to keep the polynomial generators in degree one. In this case, the polynomial ring is the cohomology of $(\mathbb{R}P^\infty)^n = B(\mathbb{Z}/2)^n$, and so the mod two Steenrod algebra acts on this polynomial ring. Again, the invariants inherit an action of the Steenrod algebra. In this case, let us look at the question of when the algebra of Dickson invariants can be the cohomology of a topological space, as an algebra over the Steenrod algebra. When $n = 1$, the group $GL(1, \mathbb{F}_2)$ is the trivial group, and the invariants form a polynomial ring on a single generator in degree one:

$$\mathbb{F}_2[c_{1,0}] = H^*(\mathbb{R}P^\infty, \mathbb{F}_2) = H^*(B\mathbb{Z}/2, \mathbb{F}_2).$$

The situation gets slightly more interesting in rank two. $GL(2, \mathbb{F}_2)$ is isomorphic to the symmetric group of degree three, and the invariants form a polynomial ring on generators in degrees two and three:

$$\mathbb{F}_2[c_{2,0}, c_{2,1}] \cong H^*(BSO(3), \mathbb{F}_2).$$

The group $SO(3)$ contains a subgroup E_2 isomorphic to $(\mathbb{Z}/2)^2$, and the normalizer quotient is $N_{SO(3)}(E_2)/E_2 \cong GL(2, \mathbb{F}_2)$. The restriction map

$$H^*(BSO(3), \mathbb{F}_2) \longrightarrow H^*(B(\mathbb{Z}/2)^2, \mathbb{F}_2)^{GL(2, \mathbb{F}_2)}$$

identifies the cohomology with the Dickson invariants.

In rank three, a similar thing happens, but with the exceptional compact Lie group G_2 . The Dickson invariants of rank three form a polynomial ring on generators in degrees four, six and seven:

$$\mathbb{F}_2[c_{3,0}, c_{3,1}, c_{3,2}] \cong H^*(BG_2, \mathbb{F}_2).$$

The group G_2 contains a subgroup E_3 isomorphic to $(\mathbb{Z}/2)^3$, and the normalizer quotient is $N_{G_2}(E_3)/E_3 \cong GL(3, \mathbb{F}_2)$. The restriction map

$$H^*(BG_2, \mathbb{F}_2) \longrightarrow H^*(B(\mathbb{Z}/2)^3, \mathbb{F}_2)^{GL(3, \mathbb{F}_2)}$$

again identifies the cohomology with the Dickson invariants.

In rank four, there is no longer a compact Lie group with the right cohomology. However, Dwyer and Wilkerson [DW2] recently constructed a space $BDI(4)$ with the Dickson invariants of rank four as its cohomology. The Dickson invariants in this case form a polynomial ring on generators in degrees eight, twelve, fourteen and fifteen:

$$\mathbb{F}_2[c_{4,0}, c_{4,1}, c_{4,2}, c_{4,3}] \cong H^*(BDI(4), \mathbb{F}_2).$$

The space $BDI(4)$ is not the classifying space of a group, but rather of a finite loop space $DI(4)$ at the prime two. We shall have more to say about this in Section 6. This finite loop space may be thought of as containing a subgroup E_4 isomorphic to $(\mathbb{Z}/2)^4$, with normalizer quotient $N_{DI(4)}(E_4)/E_4 \cong GL(4, \mathbb{F}_2)$. The “restriction map”

$$H^*(BDI(4), \mathbb{F}_2) \longrightarrow H^*(B(\mathbb{Z}/2)^4, \mathbb{F}_2)^{GL(4, \mathbb{F}_2)}$$

again identifies the cohomology with the Dickson invariants.

The game stops here. A theorem of Lin and Williams [LW] states that there is no topological space whose mod two cohomology is the ring of Dickson invariants of rank five or more (as an algebra over the Steenrod algebra).

4. Finite groups of Lie type

As we pointed out in the introduction, most finite simple groups are of Lie type. Actually, just as in the representation theory, from the point of view of the classifying space, it is better to work with certain “almost simple” groups. For example, instead of working with $PSL(n, \mathbb{F}_q)$, it is usually easier to work with $SL(n, \mathbb{F}_q)$, or even $GL(n, \mathbb{F}_q)$.

With this proviso, the data needed to specify a finite group of Lie type are a compact Lie group G , a finite field $\mathbb{F}_q$, and a diagram automorphism α of the corresponding Dynkin diagram. For a large portion of the groups of Lie type (the *untwisted* groups), α is the identity map. For example, both the finite linear groups and the finite unitary groups come from the Dynkin diagram A_n , with the identity automorphism in the first case, and a diagram automorphism of order two in the second case. The finite group associated with the above data is denoted ${}^\circ G(q)$, or just $G(q)$ if α is the identity.

The following theorem of Quillen and Friedlander (see Quillen [Nice], and Friedlander [Fr], Chapter 12) relates the classifying space of the finite group ${}^\circ G(q)$ to the classifying space of G .

Theorem 4.1. *There is a commutative diagram*

$$\begin{array}{ccc} B {}^\circ G(q) & \longrightarrow & BG \\ \downarrow & & \downarrow \Delta \\ BG & \xrightarrow{\text{id} \times (\Psi^q \circ B\alpha)} & BG \times BG \end{array}$$

which becomes a homotopy pullback square after Bousfield–Kan completing at any prime l not dividing q .

In particular, there is a fibration $B {}^\circ G(q) \rightarrow BG$ whose fibers have finite homology at primes l not dividing q .

Here, Δ denotes the diagonal map from BG to $BG \times BG$. Construction of this diagram involves replacing the compact Lie group by the corresponding algebraic group defined over $\mathbb{Z}$, and lifting from characteristic p using étale homotopy theory. The details may be found in Friedlander [Fr], Chapter 12. This diagram gives a method of calculating the cohomology of the finite groups of Lie type away from their defining characteristic. First, one uses the Eilenberg–Moore spectral sequence on this homotopy pullback to calculate the cohomology modulo a certain filtration, then one uses the subgroup structure to ungrade. This was carried out by Quillen [Qu] in the case of $GL(n, \mathbb{F}_q)$, Fiedorowicz and Priddy [FP] in the case of the other classical groups, and Kleinerman [Kl] for the exceptional Lie types.

To give an example, consider the fibration $BG_2(q) \rightarrow BG_2$ for q an odd prime power. The homology of the fiber at the prime two is an exterior algebra on generators in degrees three, five and six. So we get an embedding

$$H^*(BG_2, \mathbb{F}_2) = \mathbb{F}_2[c_{3,0}, c_{3,1}, c_{3,2}] \subseteq H^*(BG_2(q), \mathbb{F}_2).$$

The corresponding Eilenberg–Moore spectral sequence calculation gives a filtration of $H^*(BG_2(q), \mathbb{F}_2)$ whose associated graded is a tensor product of a polynomial ring on the Dickson generators in degrees four, six and seven, with an exterior algebra on generators in degrees three, five and six. After ungrading, we see that $H^*(BG_2(q), \mathbb{F}_2)$ is a free module of rank eight over the Dickson algebra. In particular, it is a Cohen–Macaulay ring, and the Poincaré series for the cohomology is

$$\sum_{i=0}^{\infty} t^i \dim_{\mathbb{F}_2} H^i(BG_2(q), \mathbb{F}_2) = \frac{(1+t^3)(1+t^5)(1+t^6)}{(1-t^4)(1-t^6)(1-t^7)}.$$

5. The Mathieu group M_{12}

A few years ago, Adem, Maginnis and Milgram computed $H^*(BM_{12}, \mathbb{F}_2)$ (see [AMM]). Their answer was rather complicated, but they observed that the answer was a finitely generated free module over a subring isomorphic to the Dickson invariants of rank three:

$$H^*(BG_2, \mathbb{F}_2) = \mathbb{F}_2[c_{3,0}, c_{3,1}, c_{3,2}] \subseteq H^*(BM_{12}, \mathbb{F}_2).$$

It is clear that there can be no homomorphism of groups from M_{12} to G_2 inducing the above map in mod two cohomology, because G_2 has a seven dimensional complex representation, while the smallest nontrivial complex representation of M_{12} is eleven dimensional. However, it turns out that there is a fibration $BM_{12} \rightarrow BG_2$ whose fibers have finite mod two cohomology. There are (at least) two possible strategies for constructing such a map. One (Milgram [Milg]) is to construct maps from BM_{12} to $BG_2(q)$ for suitable q , and then use the Quillen–Friedlander theorem. A more direct approach (Benson and Wilkerson [BW]) goes as follows. Let P be a Sylow 2-subgroup of M_{12} , of order 2^6 . Let W be the centralizer of the center of P , and W' be the normalizer of a normal fours group in P . Then $|W : P| = |W' : P| = 3$, but $W \not\cong W'$. The obvious map from the amalgamated free product $W *_P W'$ to M_{12} is a cohomology isomorphism, and so after completing at the prime two, we obtain a homotopy equivalence $(B(W *_P W'))_2^\wedge \rightarrow (BM_{12})_2^\wedge$. It turns out that there is a group homomorphism from $W *_P W'$ to G_2 , whose image is dense, and having the desired effect in cohomology. Now using Theorem 2.1, we may pass from a map between 2-completions to a map between the original classifying spaces $BM_{12} \rightarrow BG_2$ by specifying any map we want at odd primes.

The classifying space $B(W *_P W')$ can be obtained by taking the homotopy colimit (see Chapter XII of Bousfield and Kan [BK]) of the diagram

$$\begin{array}{ccc}
 BP & \longrightarrow & BW \\
 \downarrow & & \\
 BW' & &
 \end{array}$$

which may be thought of in this case as a “double mapping cylinder”.

6. Finite loop spaces

From the point of view of a homotopy theorist, the two properties of a compact Lie group G which stand out are:

- (i) G is a finite CW-complex, and
- (ii) G has the homotopy type of a loop space (namely $G \simeq \Omega BG$).

We say that a space X is a *finite loop space* if it satisfies properties (i) and (ii). It turns out that finite loop spaces enjoy many of the properties of compact Lie groups. There is an algebraically defined maximal torus, and a Weyl group which is a finite reflection group [AW].

Since homotopy theorists like to work one prime at a time using Bousfield-Kan p -completion, it is appropriate also to introduce the p -complete version of this definition. We say that X is a finite loop space at the prime p if it satisfies the following two properties:

- (i) _{p} X is p -complete and $H^*(X, \mathbb{F}_p)$ is finite, and
- (ii) _{p} X has the homotopy type of a loop space ΩBX .

Note that BX is now just the name of the space whose loops are homotopy equivalent to X . To give a finite loop space, we must really give both X and BX , but X is determined by BX so it is only necessary to construct BX .

Again, a finite loop space at the prime p has an algebraically defined maximal torus, and a Weyl group which is a finite p -adic reflection group. The finite reflection groups over the complex numbers were classified by Shephard and Todd [ST]. Clark and Ewing [CE] determined which of these were defined over the p -adics for p not dividing the group order, and demonstrated how to produce a finite loop space at the prime p in this “non-modular” case. There remains a well defined list of questions, one for each prime dividing the order of each of the groups in the list of Shephard and Todd.

There is essentially only one known example of a connected finite loop space at the prime two, which is not the 2-completion of a compact Lie group. All other known examples are given by taking products. The example in question is the space $\mathrm{DI}(4) \simeq \Omega \mathrm{BDI}(4)$ constructed by Dwyer and Wilkerson [DW2]. The Weyl group of $\mathrm{DI}(4)$ is $\mathbb{Z}/2 \times \mathrm{GL}(3, \mathbb{F}_2)$, acting as a three dimensional 2-adic reflection group. It is group number 24 in the list of Shephard and Todd. The corresponding “root system” has 21 pairs of opposite roots, so

the dimension of $DI(4)$ is $3 + 2 \times 21 = 45$. The 2-adic integral cohomology of $BDI(4)$ is a polynomial algebra on three variables in degrees 8, 12 and 28, corresponding to (twice) the degrees of the fundamental invariants of the Weyl group. The mod two cohomology is the Dickson algebra on four generators in degrees 8, 12, 14 and 15.

Dwyer and Wilkerson constructed the space $BDI(4)$ as a homotopy colimit as follows. Consider the category whose objects are the groups $(\mathbb{Z}/2)^n$ for $1 \leq n \leq 4$, and whose arrows are the injective homomorphisms between these groups. They constructed a functor to the homotopy category taking the following values:

$$\begin{aligned} \mathbb{Z}/2 &\mapsto BSpin(7)_2^\wedge & (\mathbb{Z}/2)^2 &\mapsto B(SU(2)^3 / \pm 1)_2^\wedge \\ (\mathbb{Z}/2)^3 &\mapsto B(T^3 \rtimes \mathbb{Z}/2)_2^\wedge & (\mathbb{Z}/2)^4 &\mapsto B(\mathbb{Z}/2)^4. \end{aligned}$$

They then proved that this could be lifted to a functor to the category of spaces, so that they could then take the homotopy colimit. Some of the maps involved do not exist until the spaces have been 2-completed.

The homotopy colimit of the above functor over a full subcategory consisting of a single object plays the role of the classifying space of the “normalizer” of the corresponding elementary abelian 2-subgroup in $DI(4)$. For example, the normalizer in this sense of the subgroup $(\mathbb{Z}/2)^4$ is a non-split extension of this elementary abelian subgroup by its automorphism group $GL(4, \mathbb{F}_2)$.

It is interesting to speculate on the existence of a suitable algebraic system of dimension 45 which should play the role of a tangent space at the identity for $DI(4)$. It should be almost, but not quite, a Lie algebra over the 2-adics. One tempting candidate is the set of 3×3 skew-hermitian matrices over a suitable 2-adic version of the Cayley numbers. This would have the right dimension, but on the face of it, it would seem to contain the wrong 21-dimensional Lie subalgebra (C_3 instead of B_3). But there may be some twisted version of this which works.

7. Conway’s group Co_3

Conway’s sporadic simple group Co_3 was discovered as part of a series of groups arising from Leech’s close packing of spheres in twenty-four dimensions [Co,CS]. It is defined as the stabilizer of the origin and a type three vector in the group of automorphisms of the Leech lattice, and has order $2^{10} \cdot 3^7 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$. The following theorem was proved in [Co3]:

Theorem 7.1. *There is a map from BCo_3 to $BDI(4)$, which induces an embedding*

$$H^*(BDI(4), \mathbb{F}_2) \subseteq H^*(BCo_3, \mathbb{F}_2)$$